



SEA CHANGE A rocket nose cone being

THE ISRO MODEL

FIFTY-FOUR-YEAR-OLD ISRO IS NOT ONLY THE WORLD'S MOST COST-EFFECTIVE AGENCY BUT IT IS ALSO FAST PROVING TO BE ONE OF THE MOST SUCCESSFUL INCUBATORS OF PRIVATE SPACE FIRMS

BY MANISH PANT

FROM CARRYING ROCKET CONES on bicycles and ferrying satellites on bullock carts to successfully sending an orbiter to Mars in the first attempt to making India the first spacefaring nation to land near the moon's South Pole—the Indian Space Research Organisation (Isro), the world's sixth-largest

space agency, has come a long way indeed.

In the process, the national space agency has even managed to disrupt the world of space exploration. For instance, the Mangalyaan and Chandrayaan missions were achieved on shoestring budgets of \$73 million and \$75 million, respectively, lesser than the money spent

on making Hollywood science fiction blockbusters *Gravity* (\$100 million) and *The Martian* (\$108 million).

But as is evident, Isro's frugality has its roots in its very humble origins. Just like the iconic global brands Apple and Microsoft, Vikram Sarabhai—widely hailed as the father of India's space programme—started a small space lab in the garage of his Ahmedabad residence. Starting out as the Indian National Committee for Space Research (INCOSPAR), Isro formally came into being on August 15, 1969.

Fifty-four years later, Isro is fast evolving into an agency that is enabling the creation of one of the world's biggest spacetechnology ecosystems. It has incubated and nurtured firms over those years to a point where the sector has reached critical mass and can boast of over 400 companies that are now looking to strike out on their own in activities like launch services, satellite communications, and space tourism.

And all of this has been built with